Care Creations™



Beauty Creations

Scientific Brochure

PatcH$_2$O®

The Hydra-protect technology - for Scalp & Hair

BASF
We create chemistry

PatcH$_2$O®

The Hydra-protect technology - for Scalp & Hair

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SUMMARY FILE

PatcH$_2$O®	A00297
Origin - Description	PatcH$_2$O® is a molecular patch based on film-forming natural biopolymers (hyaluronic acid, alginate and pullulan) carrying highly concentrated hydrating substances (trehalose, glycerin, urea and L-serine)
Regulatory data	
INCI	Water (and) Glycerin (and) Trehalose (and) Urea (and) L-serine (and) Pentylene Glycol (and) Glyceryl Polyacrylate (and) Algin (and) Caprylyl Glycol (and) Sodium Hyaluronate (and) Pullulan (and) Disodium Phosphate (and) Potassium Phosphate
China	Each component is listed in Inventory of Existing Cosmetic Ingredient in China (IECIC 2015)
CAS#	7732-18-5, 56-81-5, 99-20-7, 57-13-6, 56-45-1, 5343-92-0, 104365-75-5, 9005-38-3, 1117-86-8, 9067-32-7, 9057-02-7, 7558-79-4, 7778-77-0
EINECS#	231-791-2, 200-289-5, 202-739-6, 200-315-5, 200-274-3, 226-285-3, 232-680-1,214-254-7, 232-678-0, 232-945-1, 231-448-7, 231-913-4
Appearance	Colorless to yellow, clear to opalescent liquid, with characteristic odor
Preservative	None
Natural labels	None
Cosmetic uses*	
Properties	• Controlled release of hydrating complex • Film-forming effect at the surface • A barrier of protection against dehydration • Instant, extra and sustained moisture of scalp • Soothes and relax scalp • Seals the cuticle of damaged and dry hair
Applications	• Healthy scalp and Beautiful hair - For all type of hair • Scalp & hair sustained moisturization, post-hair dye/straightening soothing serum, hydra-comfort sulfate free shampoo, healthy look conditioner, ultra-moisture conditioner for ethnic care • Suitable for neutral and soft formula • Water soluble – non greasy – easy to wash
Formulation data	
Concentration of use	2 to 3%
Solubility	Soluble in water
Incorporation method	Incorporate into the aqueous phase during the formulation process up to 85°
Optimal pH	4 - 8
Storage	10-30°C, protect against heat and from the effects of light
Shelf life	24 months
Patent family	Application filed

® = trademark of BASF in several countries
* Marketing and scientific brochures for skin care are also available



THE KEYS TO HYDRATION FOR HEALTHY SCALP AND BEAUTIFUL HAIR

Water Source of all life

Water is the origin of life. Water is essential for the human body. People are fully aware of the importance of a natural water equilibrium in their body to maintain their health and beauty. For the cosmetic market, hydration still represent the second largest claim sought by consumers. We naturally thought to protect our skin from dehydration but what about scalp and hair protection? **Too often overlooked, the scalp is however a natural reservoir of life for healthy hair and required the same care than the rest of the body**.

Superficial but not superfluous

Talking about **skin hydration** is above all talking about hydration of the upper layer of the epidermis or *stratum corneum*. Under physiological conditions, this is normally ensured by the **natural moisturizing factors** (NMFs)[1].

Thanks to the complex blend of amino acids, minerals and osmolytes (Table 1), NMFs have a phenomenal water retention capacity. On their own, NMFs ensures half of the water retaining capacities of the *stratum corneum*[2]. NMFs act at the surface of the skin, because most of its components are found no more than 60 µm from the surface[3]. NMFs are obviously not the only natural hydration factors. Hyaluronic acid (HA) has also a role through its hygroscopic power[4].

Such as skin, **scalp and hair are daily aggressed by environmental factors** as sun, wind, chlorine, saltwater or dry air. Habits such as over-washing, brushing, thermal abuse or chemical treatments further weaken the hair fiber causing the water to evaporate leading to dryness. **The scalp become uncomfortable and itchy[5,6] and the hair fibers lose their body, shine, and vitality. Hydration is a key to preserve and restore hair beauty**.

Challenges and strategies

Cosmetics have mostly used a cocktail of moisturizing ingredients or Hyaluronic acid to mimic the natural moisturizing factors for decades to bring hydration to the skin.

Topical application of NMFs reduces water loss and increases the capacity of the skin to retain moisture within half an hour following application[7]. This is also true in case of the scalp moisturization. A well moisturized scalp is more comfortable, relaxed, and healthier.

Improving the hydration of the scalp also impacts the quality of hair.

The main challenge is clearly to achieve immediate hydration of the scalp along with long lasting efficacy. In general, the effectiveness of the components of NMF applied topically is certainly quick but may not be long lasting. Therefore, it is important to increase the bioavailability of the hydrating components by slowing down their diffusion by controlling their release. Secondly it is also important to improve the skin barrier to fight against water evaporation.

A widely experienced expert in the vectorization and controlled release of active cosmetic ingredients, BASF Beauty Care Solutions, offers an efficient technical solution - PatchH_2O® to meet the needs to restore hydration and to form a protective film to avoid loss of water from the scalp and from hair.

Table 1 - Chemical composition of natural moisturizing factor[8]

Chemical	Composition (%)
Free amino acids	40
Pyrrolidone carboxylic acid	12
Lactate	12
Sugars	8.5
Urea	7
Chloride	6
Sodium	5
Potassium	4
Ammonia, uric acid, glucosamine	1.5
Calcium	1.5
Magnesium	1.5
Phosphate	0.5
Citrate and formate	0.5



UNCOMFORTABLE SCALP AND DRY HAIR

Dry scalp, just like skin is mainly perceived as itchy and uncomfortable[9]. It is caused by dehydration resulting in scalp irritation causing people to scratch.

Many people around the world is concerned by this scalp trouble, though the severity can vary quite widely. In France, about 25% of people has dry scalp and 44% suffer from itch[9]. In China, about 57% of people suffer from these concerns[6]. African-descendent people also exhibit relatively dry and uncomfortable scalp due to naturally irregular distribution of sebum[10] that lubricates the scalp.

Dry and itchy scalp depends on ambient factors, such as dry climate, colder temperatures, and repeated washing[9]. With increasing deleterious lifestyle, pollution, emotional stress and nutritional misbalance, the scalp troubles are increasing. **Dry scalp is a very common problem in modern life, affecting people from different industrial areas and diverse age groups**. Aggressive treatments such as dyeing[11], bleaching, perming or straightening tend to weaken the scalp and affects its ability to keep moisture leading to an increasing sensitivity.

To correct the scalp concerns, it is often advised to use gentle products and to massage and stimulate the scalp. Scalp hydration is important and highly recommended after sunbath or swimming. In fact, scalp needs hydration and care just like the skin. So far, scalp concerns have been neglected, but it is now clear that special care is needed for revealing the full beauty potential of hair. **In fact, the secret of beautiful hair lies at its root within healthy scalp**.

Dry scalp leads to dry hair. Hair grooming as well as daily washing, heating, drying[12] and brushing increases its perceived dryness and fragility. Handling it roughly causes split ends and hair breakage. In addition, hair treatments such as dyeing, straightening, curling can induce damages which increase the loss of water from hair. Kinky and wavy hair from African or Hispanic ethnicities are particularly prone to hair breakage[10]. **Dry hair appears brittle, looks lifeless and is difficult to brush. Consumers have the desire to treat dry hair with moisturizing products to recover its full body, shininess, and strength**.

THE HYDRA-PROTECT CONCEPT TO FIGHT AGAINST SCALP AND HAIR DEHYDRATION

The hydra-protect concept

PatchH_2O®, BASF's skin moisturizer, is based on highly advanced liquid patch technology for the controlled release of moisturizing factors. Its unique combination of pullulan, alginate, and hyaluronic acid, forms a molecular network that acts as a protective film and reduces water evaporation on the skin's surface.The micromesh contains a complex made up of glycerin, L-serine, trehalose, and urea. These natural moisturizing factors are gradually released into the upper layer of the epidermis – the *stratum corneum* – rebalancing its moisture levels. The 'anti-thirst' ingredient's efficacy has been demonstrated in various *in vitro* and *in vivo* studies[13].

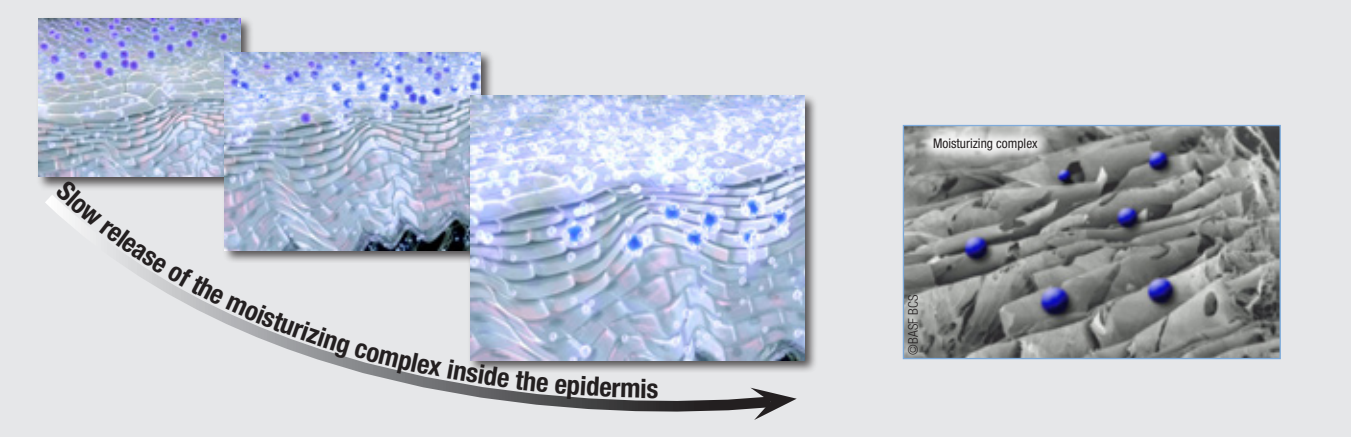


Figure 1 - The Hydra-protect technology of PatchH_2O® for sustained skin or scalp hydration.

Rapid and sustained effectiveness

Although one out of three treatments on the market makes hydration claims, very few products efficiently give the skin both immediate and sustained hydration. PatchH_2O® offers this effectiveness thanks to the combination of the two areas of BASF's expertise - natural macromolecular and controlled release technology (Figure 1).

To fight against scalp and hair dehydration

PatchH_2O® was developed to fight against skin dryness and to ensure a sustained hydration. In addition to skin care applications, BASF Beauty Care Solutions' team has validated by in vivo study that **PatchH_2O® can also be used in scalp care applications, especially to consistently hydrate, soothe and calm dry and uncomfortable scalp. PatchH_2O® also displays other hair benefits**. It rebalanced the hair water content and brought perceivable improvement of hair quality and shine. This makes PatchH_2O® suitable for anti-aging scalp and hair products.



THE HYDRA-PROTECT TECHNOLOGY:
A MICROPATCH OF 3 NATURAL BIOPOLYMERS AND A MOISTURIZING COMPLEX

A 3D meshwork of 3 natural biopolymers: Pullulan, Alginate and Hyaluronic acid

PatcH$_2$O® is a meshwork of 3 natural bio-polymers which allows sustained release of the moisturizing complex. Thanks to the combination of pullulan, sodium alginate and hyaluronic acid, the 3D meshwork of PatcH$_2$O® technology gives a new youthfulness to classic moisturizing ingredients like glycerin, urea, L-serine and trehalose (Figure 2). PatcH$_2$O® has a double function: it acts both as a protective film at the surface and as a reservoir for water[13].

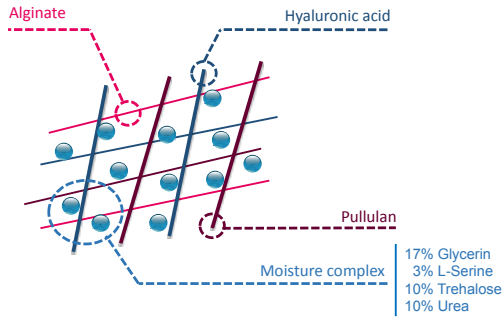


Figure 2 - Scheme of PatcH$_2$O® components: micropatch (hyaluronic acid, alginate and pullulan) and moisturizing complex (glycerin, L-serine, trehalose, urea).

A well-organized 3D micropatch of natural biopolymers

The molecular mesh in PatcH$_2$O® is a combination of three polysaccharides - pullulan, alginate, and hyaluronic acid in 1:2:1 proportion, respectively. The well-organized 3D structure is enabled by the establishment of hydrogen bonds (Figure 3).

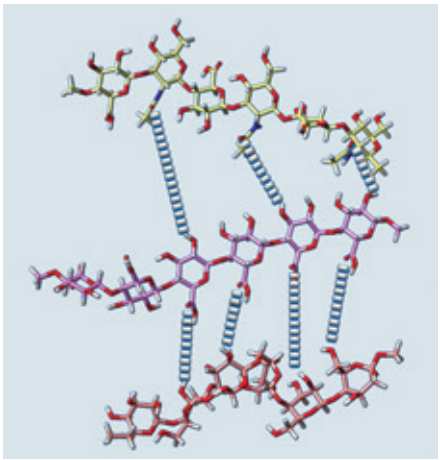


Figure 3 - 3D modeling of the "Hydrogen" interactions between hyaluronic acid, alginate and pullulan.

The efficient combination of the three biopolymers forms a film-like structure with a multi-layered organization that is easily observed after dehydration using imagery microscopic techniques. (Figure 4A). This structure is also clearly visible in scanning electron microscopy (Figure 4B).

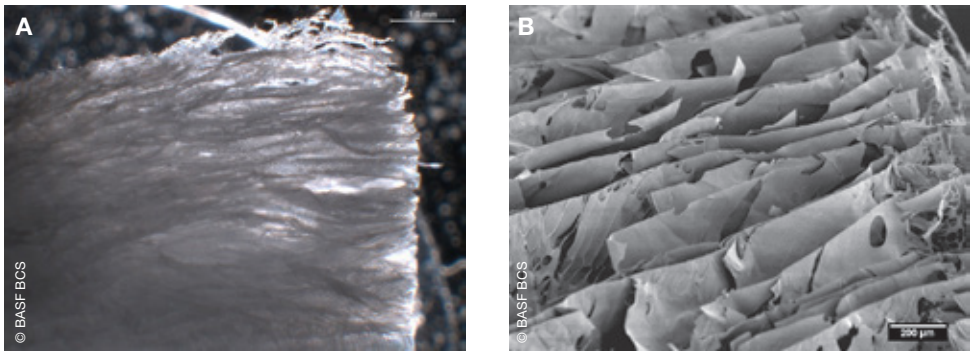
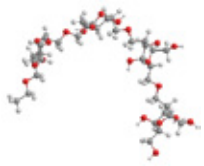


Figure 4 - Microscopic observations of PatchH_2O® meshwork: **A** - Bright-field by macroscopy (scale bar: 1 mm)
B - Scanning electron microscopy (scale bar: 200 µm)

Three natural biopolymers

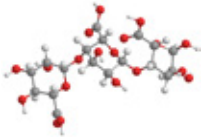
The three polysaccharides - pullulan, alginate and hyaluronic acid have been chosen for their physiochemical properties.

Pullulan



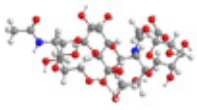
Pullulan is a linear polysaccharide composed of repeating maltotriose units (poly-α-1,6-maltotriose). The polymeric structure gives unique gelling, adhesive and film forming properties[14]. Pullulan is a natural ingredient produced by the fungal cells of Aureobasidium pullulans through a bio-fermentation process of a food-grade starch. Pullulan has been sold as a food additive in Japan for more than 20 years, and the FDA has certified it as a safe food ingredient[15]. Pullulan also has antioxidant benefits that, in combination with collagen, can increase the survival of stem cells placed in high oxidative stress and maintain them longer in their pluripotent state[16].

Alginate



Alginic acid is a polymer of D-mannuronic and L-glucuronic units obtained from brown algae (Phaeophyceae) by basic hydrolysis. Long used as bulking agents in pharmaceuticals for their gelling, stabilizing and film-forming qualities, alginate gels have an ideal porosity that allows them to trap small molecules while maintaining their free diffusion inside and outside a gel. Associated with certain drugs, alginate regulates the rate and kinetics of absorption of active ingredients[17].

Hyaluronic acid



In recent years, hyaluronic acid (HA) has emerged as the most popular category of injectable wrinkle fillers as anti-aging substance[18]. Despite of this, it is also used as moisturizer for its dual properties as humectant and emollient. Half of the epidermal HA can be found in the *stratum corneum*. This hygroscopic molecule can hold up to 1,000 times its volume in water[19]. In addition to its ability to attract water in *stratum corneum*, it can also fill in spaces between corneocytes in interaction with other molecules such as phospholipids, to form complex networks[20].

A moisturizing complex

The complex of hydrating factors

PatcH$_2$O® would be nothing without the moisturizing complex that it contains: glycerin, L-serine, urea and trehalose.

L-serine and urea

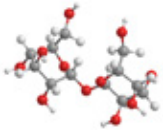


L-serine and urea are both components of NMF[1]. The free hydroxyl group of L-serine enable binding to water molecules to form stable complexes[21].



Urea is a potent osmolyte actively involved in the regulation of the salinity of the extracellular environment[22].

Trehalose



Trehalose stabilizes cell membranes and preserves their integrity by binding to phospholipids, maintaining their spacing, by forming a protective layer on the surface and sometimes even by taking the place of water molecules[23]. Trehalose provides hair with heat styling benefits in high humidity environment[24].

Glycerin

Glycerin is a common humectant often used as skin[25] and hair moisturizer. It is able to absorb humidity from the atmosphere as well as from within the skin. It is often used by people with extremely dry skin or hair[26].

A safe product suitable for scalp and hair care

Safety/tolerability of the product

PatcH$_2$O® was tested to ensure its safety in the recommended dose of use. PatcH$_2$O® does not irritate eyes or skin and no indication of skin sensitization and photo-toxicity was observed.

A product suitable for gentle formula

PatcH$_2$O® is a colorless fluid with a composition **specifically chosen for non-ionic formula**. PatcH$_2$O® provides formula with a silky smooth, non-greasy feel and leaves a sensation of comfort on the skin/scalp. PatcH$_2$O® is water-soluble, non-greasy, and easy to remove by rinsing. PatcH$_2$O® does not contain any preservatives.

Compatibility notes: product may cause incompatibilities with cationic components and ionic surfactants.

Description

PatcH$_2$O® is an aqueous solution based on the innovative association of film-forming polymers and a powerful hydrating complex.

INCI name: Water (and) Glycerin (and) Trehalose (and) Urea (and) L-serine (and) Pentylene Glycol (and) Glyceryl Polyacrylate (and) Algin (and) Caprylyl Glycol (and) Sodium Hyaluronate (and) Pullulan (and) Disodium Phosphate (and) Potassium Phosphate

Dose of use: 2 - 3%

Example: commercial sample of PatcH$_2$O® (2), incorporated respectively into a hair lotion at 2% (1) or an emulsion (3) or a hydrogel (4)



China IECIC listed



A DEMONSTRATED EFFECTIVENESS TO FIGHT AGAINST SCALP AND HAIR DEHYDRATION

While PatchH_2O® is unique in its composition, the innovation lies in its performance: *in vitro*, *ex vivo* and *in vivo*.

PatchH_2O® rebalances water content in scalp by an immediate, cumulative and sustained moisturizing effect with proven efficacy. Use of PatchH_2O® containing formulations resulted in hydrated and more comfortable scalp as well as imparted a healthy attribute to hair.

PatchH_2O® developed by Beauty Creations perfectly meets the requirements of the moisturizing cosmetic as well as growing scalp care market. It is particularly suitable for a **healthy scalp** and **beautiful hair** market segments.

EFFICACY FOR SCALP CARE

Water retention – Vitro-skin® model

ex vivo

OBJECTIVE

The purpose of this study was to evaluate the ability of PatcH$_2$O® to retain water in the *stratum corneum* by an accelerated evaporation test, using synthetic skins (Vitro-skin®).

RESULTS & DISCUSSION

PatcH$_2$O® at 3% increased the retention of water by +19.7% in Vitro-skin® at 48 hours (Figure 5). Vitro-skin® treated with PatcH$_2$O® at 3% retained significantly more water than artificial skin treated only with the hydrating complex (control without bio-polymers) at each time point. This water retention level increased up to a factor x3 after 48 hours comparatively to the hydrating complex alone (p<0.001).

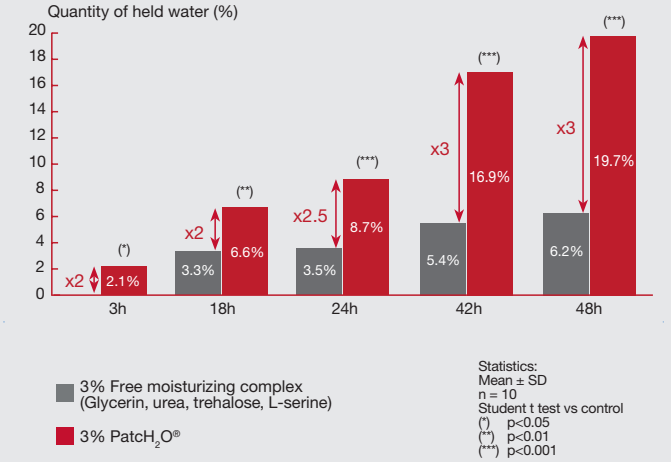


Figure 5 - Comparison between PatcH$_2$O® and the hydrating complex alone to limit the evaporation of water on synthetic skin (Vitro-skin®).

CONCLUSION

PatcH$_2$O® promotes the maintenance of the water molecules in the skin for 48 hours (Vitro-skin®) thanks to its unique combination of bio-polymers and its film-forming properties. This assay demonstrates the capacity of PatcH$_2$O® to offer long-lasting scalp's surface hydration.

MATERIALS&METHODS

Material

Vitro-skin® is a usual mimetic artificial skin substrate for *ex vivo* tests. Vitro-skin® exhibits similar characteristics to human skin's surface, such as topographies, pH, ionic strength and critical surface tension *(http://www.ims-usa.com/vitro-skin)*.

Treatment

PatcH$_2$O® at 3% or moisturizing complex only (without bio-polymers) and deionized water (control) are applied on 3 cm² squares of Vitro-skin®. The treated skins are then dried at 45° C in a thermo-regulated oven for 1 hour to obtain a cohesive film on the surface.

Water retention measurement[27]

Dried skins were positioned on containers containing approximately 6 cm^3 water, then placed again in the oven at 45°C for 48 hours of evaporation. Water retention measurement was achieved by measuring the amount of water evaporated over the time. Each container's water loss was then measured by regular weighting. Water retention ability of PatcH$_2$O® was calculated in comparison with the moisturizing complex alone. Results are expressed as a percentage of water maintained in comparison to the untreated control.

Statistics

10 artificial skins were processed for each tested formula and expressed as mean value ± standard deviation (SD). Statistical analysis was performed using Student t test, with a threshold of significance set at 5% (p<0.05).

EFFICACY FOR SCALP CARE

Controlled release of moisturizing agents – human *stratum corneum*

ex vivo

OBJECTIVE

The aim of this study was to evaluate the controlled release property of PatcH$_2$O®. The diffusion and storage of urea or L-serine contained in PatcH$_2$O® were quantified in the *stratum corneum* isolated from human skin explants and compared with those of the free active ingredients alone (control without bio-polymers).

RESULTS & DISCUSSION

Storage of L-serine in the *stratum corneum*

Storage of L-serine in the *stratum corneum* was significantly increased in human skin treated with PatcH$_2$O® compared to free L-serine control at all timepoints tested (Figure 6).

Storage of urea in the *stratum corneum*

Storage of urea in the *stratum corneum* was significantly increased in skin treated with PatcH$_2$O® compared to the free urea control at all timepoints tested (Figure 7).

Controlled release of L-serine and urea by the meshwork of bio-polymers

PatcH$_2$O® significantly increased the storage of urea and L-serine in the *stratum corneum* at least 48 hours after application compared to the hydrating molecules alone.

This study highlights the ability of PatcH$_2$O® to control the release of hydrating molecules into the *stratum corneum* thanks to the meshwork of bio-polymers.

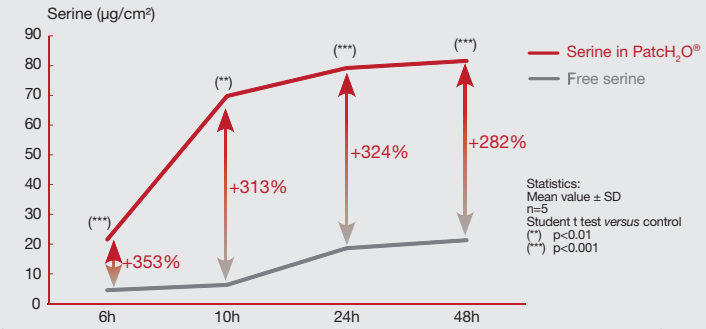


Figure 6 - Serine diffusion and storage in *stratum corneum*.

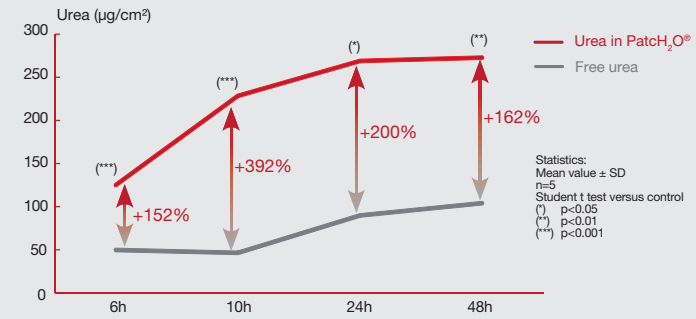


Figure 7 - Urea diffusion and storage in *stratum corneum*.

CONCLUSION

PatcH$_2$O® Hydra-protect technology allows long-term maintenance of L-serine and urea in the superficial part of the epidermis thanks to the controlled release of the hydrating complex.

MATERIALS&METHODS

Cell culture

Human skin explants were placed on Franz diffusion cells and equilibrated for 14 hours with phosphate buffered saline (PBS) containing antibiotics. 300 µl of the products to be tested were then added to the "donor" compartments of the cells, PBS solution was then added to the "receptor" compartments.

Treatment

Skin explants were incubated in the presence of free L-serine or free urea (controls, final concentration of 3% and 10% respectively) and PatcH$_2$O® containing active ingredients at the same concentration. 300 µl of PatcH$_2$O® and moisturizing complex were applied to human skin biopsies. Treated skins were placed on Franz cells and incubated for 48 hours at 37°C.

L-serine and urea measurements

At various times (6, 10, 24 and 48 hours), skin biopsies were recovered, stripped (D-Squam®, CuDerm) and quantities of L-serine and urea present in the *stratum corneum* were measured using HPLC. All experiments were performed in quintuplicate (n = 5).

Values and statistics

Results expressed in mg of L-serine and urea/cm², are given in the form of the mean ± SD. Statistical significance was assessed using Student t test with a threshold of significance set to 5% (p<0.05).



Franz cell

EFFICACY FOR SCALP CARE

Measurement of skin hydration – human *stratum corneum*

ex vivo

OBJECTIVE

The purpose of this study was to demonstrate the effect of PatcH$_2$O® in epidermal hydration. Dielectric conductivity of *stratum corneum* from human skin biopsies treated with 3% of PatcH$_2$O® was compared with a skin treated with 3% moisturizing complex alone.

RESULTS & DISCUSSION

PatcH$_2$O® formulated at 3% in a standard hydrogel increased the water content from 30 min to 48 hours compared to the baseline (before application) (Figure 8). The water content increased by 129% at 4 hours after application of PatcH$_2$O®, and by 106% at 48 hours.

Stratum corneum hydration was significantly increased by 1.6 at 48 hours after treatment with PatcH$_2$O®, compared to the moisturizing complex alone.

The hydrating effect of PatcH$_2$O® in a standard formulation has been demonstrated on human skin explants. PatcH$_2$O® at 3% offered a more intense and immediate moisturization to the upper layers of the epidermis (30 minutes and 4 hours after the treatment) and a lasting effect (up to 48 hours) compared to the hydrating complex alone.

The results of this study confirm the long-term benefit to use the polymer network of PatcH$_2$O®.

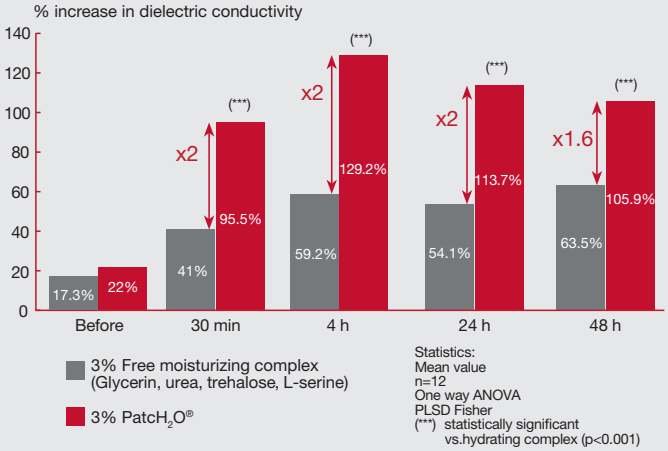


Figure 8 - Hydrating effect of PatcH$_2$O® compared to the moisturizing complex alone from 30 min to 48h after treatment in human *stratum corneum*.

CONCLUSION

PatcH$_2$O® Hydra-protect technology allows maintenance of hydrating ingredients in superficial part of the epidermis to support immediate and long lasting hydration.

MATERIALS&METHODS

Stratum corneum preparation

This study was conducted with human skin explants from plastic surgery. Epidermal sheets were prepared by peeling of the dermis after heating at 60° C for 2 minutes. The *stratum corneum* was isolated by enzymatic digestion as described[28].

Treatment

Epidermal sheets were balanced in an oven set at a relative humidity (44% - saturated solution of potassium carbonate), then were treated with a placebo (control hydrogel without active ingredients), a control hydrogel containing 3% of the moisturizing complex alone and hydrogel containing PatcH$_2$O® at 3%. 1 mg/cm² of each formula was applied consecutively three times on *stratum corneum* with an interval of 30 minutes (12 measurements).

Dielectric conductivity measurement - Tagami test[29,30]

Water content on *stratum corneum* were measured by the electrical conductivity method using a hygrometer Skicon 200 (IBS Inc., Hamamatsu, Japan). Values are expressed in micro-Siemens (µS). The principle of the method is based on the relationship between variation of skin electrical conductivity and its water content, due to the bipolarity of water molecules. The more the *stratum corneum* is hydrated, the higher conductivity is. Dielectric conductivity measurements were carried out prior to treatment and then up to 48 hours after the third application. The results of PatcH$_2$O® and the hydrating complex alone was therefore expressed in percentage of increase in dielectric conductivity compared to placebo (hydrogel alone).

Results and statistics

Measurements were performed in triplicates on each sample and for each time interval.

Statistical analysis was performed using one-way ANOVA followed by PLSD Fisher test with a threshold of significance set at 5% ($p<0.05$).

Conclusion

ex vivo

PatcH$_2$O® on human scalp

Hydra-protection for scalp

- Controlled release of hydrating factors: at least 48 hours (human *stratum corneum*).
- Film-forming effect against dehydration: + 19.7% of water retention at 48 hours (Vitro-skin®).
- Immediate and long lasting hydration: + 129% at 4 hours and + 106% at 48 hours (human *stratum corneum*).

Advantages of the 3D meshwork of bio-polymers over the hydrating complex alone

- 3x more water retention at 48 hours (Vitro-skin®).
- + 282% of L-serine and + 162% of urea storage at 48 hours (human *stratum corneum*).
- 2x more hydration at 4 hours and 1.6x more at 48 hours (human *stratum corneum*).

PatcH$_2$O® provide increasing moisture and a film-forming protection which act as a barrier against dehydration of human sclap.

EFFICACY FOR HAIR CARE

Illustration of the film-forming effect – hair tresses

ex vivo

OBJECTIVE

The purpose of this study was to image the film-forming effect of PatcH$_2$O® meshwork on the surface of damaged hair fiber.

RESULTS & DISCUSSION

Thermal damaged hair was treated with PatcH$_2$O® at 2% in a leave-on solution. Single hair fiber treated with PatcH$_2$O® is shown in Figure 9. For comparison, a part of the fiber not exposed to PatcH$_2$O® is also illustrated.

PatcH$_2$O® at 2% in the leave-on aqueous solution formed a well-defined film on the hair surface (Figure 9).

Additional experiments using hair tresses moderately damaged by oxidative bleaching and treated with a leave-on solution of PatcH$_2$O® were also performed, and provided similar results (picture not shown).

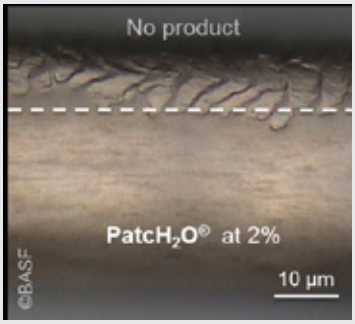


Figure 9 - Imagery of a single hair fiber covered by the film-forming structure of PatcH$_2$O® (at the bottom of the picture) and uncovered (at the top of the picture) by microscopy (Hirox camera).

CONCLUSION

The PatcH$_2$O® meshwork, composed by the natural bio-polymers and the moisturizing complex applied on damaged hair, is able to form a film-like structure such as a barrier covering the hair surface.

MATERIALS&METHODS

Hair tresses

European normal human hair strands were washed and treated with a hair straightener to perform medium thermal damages on hair fibers (10 min at 200°C). Single hair fibers were fixed in a glass slide and covered by PatcH$_2$O® at 2% in the leave-on aqueous solution. Samples were dried overnight at room temperature in an air-conditioned room chamber.

Microscopy

Single hair fibers were imaged by a 3D digital video microscope (KH-8700, Hirox Europe Ldt.)

EFFICACY FOR HAIR CARE

Protection of damaged hair against dryness - hair tresses

ex vivo

OBJECTIVE

The purpose of this test was to evaluate the protective effect of PatchH_2O® against dehydration of damaged hair as a function of time. The hair water content was monitored by Corneometer® measurements after the treatment with an aqueous solution and during the natural drying of hair strands. Untreated virgin hair was taken as reference for water content in healthy hair. Bleached hair treated with water only was taken as control.

RESULTS & DISCUSSION

Kinetics of hair water content from 0 to 24h after the aqueous treatment allowed us to monitor the modulating effect of PatchH_2O® on water evaporation in damaged hair. Virgin hair and damaged hair treated with PatchH_2O® were compared to damaged hair.

Immediately after the treatment with the aqueous solution the level of water content is maximum (T0). Virgin hair was less hydrated compared to damaged hair. No significant difference was observed between untreated and treated damaged conditions (Figure 10A).

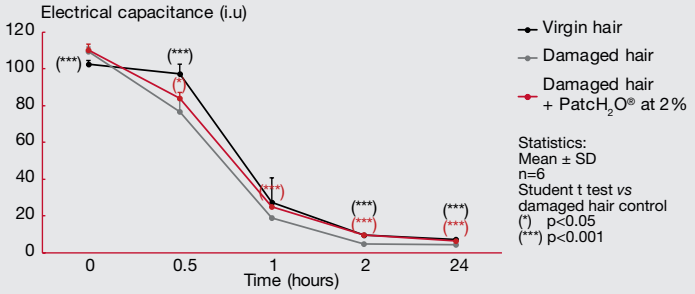
From 30 min to 24h after the treatment and as expected, bleached hair exhibited less water content than virgin hair due to the higher water evaporation from the damaged cuticle.

PatchH_2O® at 2% has significantly increased the water content in damaged hair compared to the virgin hair control after 2 hours (Figure 10A). It is important to note that no significant difference was measured in water content from damaged hair treated with PatchH_2O® at 2% and virgin hair from 1h to 4h after the treatment (data not shown).

Data were then normalized to virgin hair (as reference of healthy hair) and damaged hair (at baseline, as reference of dry hair). PatchH_2O® at 2% has restored by 97% the water content of damaged hair at 2 hours (Figure 10B).

A

Kinetics of water loss in virgin and damaged hair



B

Potential to restore water balance in dry damaged hair



Figure 10 - Protection of damaged hair against dryness: **A** - Kinetics of hair water content in virgin hair and damaged hair treated with PatchH_2O® at 2% compared to damaged hair (medium bleaching) during the natural drying of hair (at room temperature). **B** - Potential of PatchH_2O® to restore water balance in dry damaged hair.

CONCLUSION

PatchH_2O® imparts protection of damaged hair against dryness in a short and long-term manner. PatchH_2O® restores the hair water balance of dry hair to be close to that of healthy virgin hair.

MATERIALS&METHODS

Hair tresses

European normal human hair tresses were washed with a standard shampoo, rinsed and dried carefully before experiments. One part of the normal hair strand was taken as virgin hair. The second part was medium bleached with an oxidizing mixture and rinsed, washed, and dried overnight in controlled climatic conditions. A part of the medium bleached tress was taken for the untreated damaged hair condition, while the resting part of the damaged hair were treated with PatchH_2O® at 2% in a leave-on aqueous solution. Controlled hair conditions were treated with the water only. Hair tresses were dried in an air -conditioned room at a temperature of 21± 1°C and at 50 ± 5% relative humidity.

Corneometer® measurements

Water content in hair were measured by the electrical capacitance method with the Corneometer CM 825 (Courage & Khazaka, Köln, Germany). The principle of measurement is based on the correlation between di-electricity values and water content. Corneometer® values are the mean of 5 measurements per tress.

Results and statistics

Results are expressed as mean ± SD for 6 independent tresses per condition. Statistical comparison of data between conditions was performed by the two-sided student t test. Differences between values were considered significant for p-value <0.05. Data were normalized to virgin hair and damaged hair to evaluate the potential of the product to restore water content of damaged hair.

EFFICACY FOR HAIR CARE

Hair surface repair – hair tresses

ex vivo

OBJECTIVE

The aim of the study was to assess the repair effect of PatcH$_2$O® on damaged hair surface by scanning electronic microscopy (SEM). Measurements were performed on medium bleached human hair tresses after application of the PatcH$_2$O® at 2% in a leave-on aqueous solution. Untreated virgin hair was taken as reference of healthy hair. Bleached hair treated with water only was taken as control.

RESULTS & DISCUSSION

Damage of hair cuticle was observed by the increasing number of spiky flakes of detached scales compared to virgin hair surface (Figure 11A). The percentage of detached scales quantified by image analysis increased significantly in medium-bleached hair compared to virgin hair (Figure 11B; $p<0.001$).

PatcH$_2$O® at 2% restored hair surface quality of damaged hair in comparison with untreated damaged condition (Figure 11A). The percentage of spiky flakes from detached scales of the cuticle decreased significantly by -37% in damaged hair treated by PatcH$_2$O® at 2% in comparison to the untreated damaged condition (Figure 11B, $p<0.001$).

PatcH$_2$O® at 2% significantly restored the hair cuticle quality of oxidatively damaged hair by 83% (Figure 11C), reverting to a surface quality similar to that of virgin hair.

CONCLUSION

Treatment of damaged hair by PatcH$_2$O® restores the hair surface close to quality of the virgin hair surface. Using PatcH$_2$O®, scales are sealed.

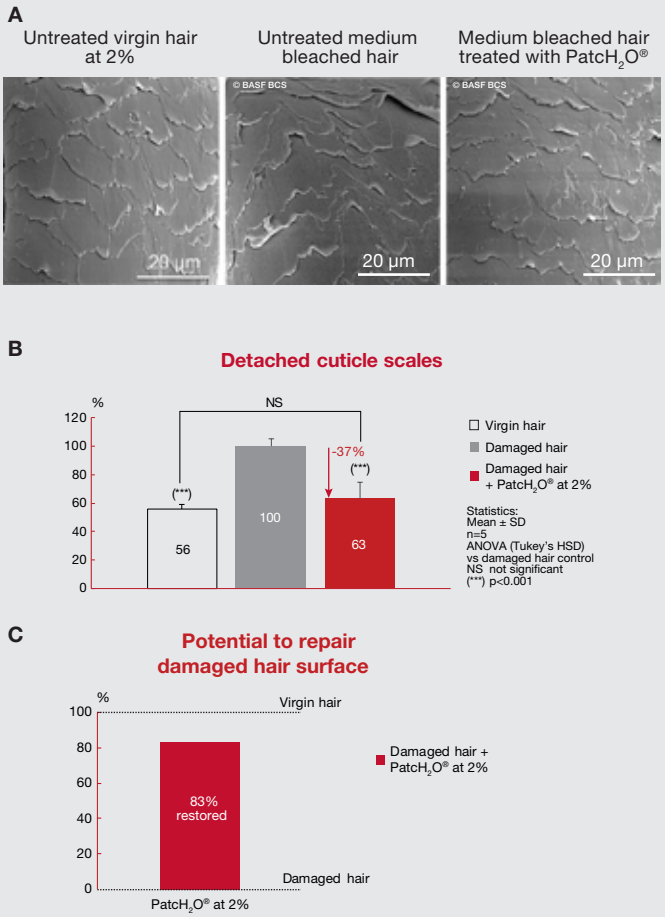


Figure 11 - Hair surface repair. **A** - Images of hair surface by scanning electronic microscopy. Scale bar: 20 µm. **B** - Quantity of spiky flakes on human hair surface represented by the detached/total flakes ratio (%) by image. **C** - Repair effect of PatcH$_2$O® at 2% on damaged hair cuticle.

MATERIALS&METHODS

Hair tresses

European normal human hair tresses were washed with a standard shampoo, rinsed and dried carefully before experiments. One part of the normal hair strand was taken as virgin hair. The second part was medium bleached with an oxidizing mixture, then rinsed and dried overnight in controlled climatic conditions. A part of the medium bleached tress was taken as untreated damaged hair condition, while the other part of the damaged hair was treated with PatcH$_2$O® at 2% in an aqueous leave-on solution. Control hair conditions were treated with the water only.

Scanning electron microscopy (SEM)

5 fibers were taken per conditions, shaped into a node and prepared for electronic microscopy imagery in a standardized procedure.

Image analysis

SEM images were analyzed to evaluate hair surface damages by counting of detached cuticle flakes in two areas of interest. Number of detached flakes was normalized by total flakes and the mean of values were expressed as a percentage.

Statistics

ANOVA with factor treatment was performed, including pairwise comparisons of products for assessment time and application (Tukey's HSD). Statistical significance level of $\alpha = 0.05$ was chosen for statistical analysis.

ex vivo

Conclusion

PatcH$_2$O® on human dry hair

Hair damaged by chemical processes or heating are considered as dry hair. Our experiments confirmed that hair damaged by chemical oxidation exhibited a lower content of water in hair and an increased number of detached flakes on the cuticle compared to the virgin hair.

The film formed by PatcH$_2$O® was visible at the surface of damaged hair after treatment with the leave-on solution at 2%. Treatment of damaged hair with PatcH$_2$O® at 2% increased the quality of hair surface by decreasing the number of detached flakes and consequently by increasing the water content in hair.

Hair benefits using PatcH$_2$O®

- A visible film-forming effect on hair surface after aqueous leave-on treatment.
- A protection of damaged hair against dryness.
- A restoration of hair water content up to 97% in 2 hours.
- A decrease by -37% of detached flakes on hair cuticle.
- A repair by 83% of hair cuticle.

PatcH$_2$O® at 2% provides a film-forming protection such as a barrier against dryness of damaged hair and re-balances the water content of dry hair in a similar manner to that of healthy hair.

CLINICAL EFFICACY AND CONSUMER PERCEPTION

Hydra-protect effects of Patch$_2$O® on scalp and hair – human volunteers

in vivo

OBJECTIVE

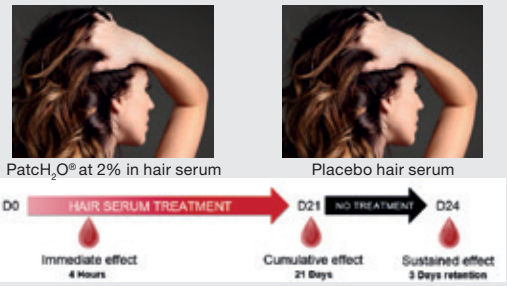
The aim of the study was to evaluated the hydra-protect effects of PatcH$_2$O® on dry and uncomfortable scalp, and on dry hair. A double-blind placebo-controlled clinical study was performed on 65 volunteers divided in 2 groups to test the placebo or the PatcH$_2$O® containing product.

Volunteers from multi-ethnic origin (Caucasian, Afro-American, Hispanic and Asian) applied and massaged a hair serum containing PatcH$_2$O® at 2% or the placebo formula on the whole scalp once daily for 21 days.

Scalp hydration was evaluated by electrical conductance (Skicon) at baseline, 4 hours, 21 days, and 3 days after the end of the treatment (3-day regression period) to evaluate the immediate, cumulative and sustained effect of the product respectively (Figure 12).

Consumer self-perception of the product effectiveness on scalp and hair was assessed by a questionnaire after 21-day treatment and after 3-day regression.

65 women (Caucasian, Afro-American, Hispanic and Asian) having dry and itching/uncomfortable scalp, and dry hair, divided in two groups.
Double-blind, randomized study. Once daily application during 21 days.



Measurement of scalp hydration by conductance (Skicon) at D0, 4h, D21 and 3 days after the last application.
Consumer self-perception on scalp comfort and hair quality benefits at D21 and 3 days after the last application.

Figure 12 - Scalp hydration by clinical study. Patch$_2$O® *vs* Placebo. Protocol.

RESULTS & DISCUSSION

Immediate and cumulative hydrating effect (during the treatment)

Immediate effect: PatcH$_2$O® significantly increased by 17% the scalp hydration at 4 hours after the first application ($p<0.01$), whereas the placebo did not display any significant effect (Figure 13A).

Cumulative effect: PatcH$_2$O® significantly increased scalp hydration by 52% over baseline ($p<0.001$), and by 36% over placebo ($p<0.01$) after 21 days of daily application.

Sustained effect (3 days after stopping the treatment)

After 3 days without any product application, PatcH$_2$O® prevented the dehydration of the scalp, whereas scalp hydration significantly decreased by -18% over baseline in the placebo ($p<0.05$) (Figure 13B). This prolonged hydrating effect on the scalp after stopping the treatment is significant versus the placebo group (+ 27%, $p<0.05$).

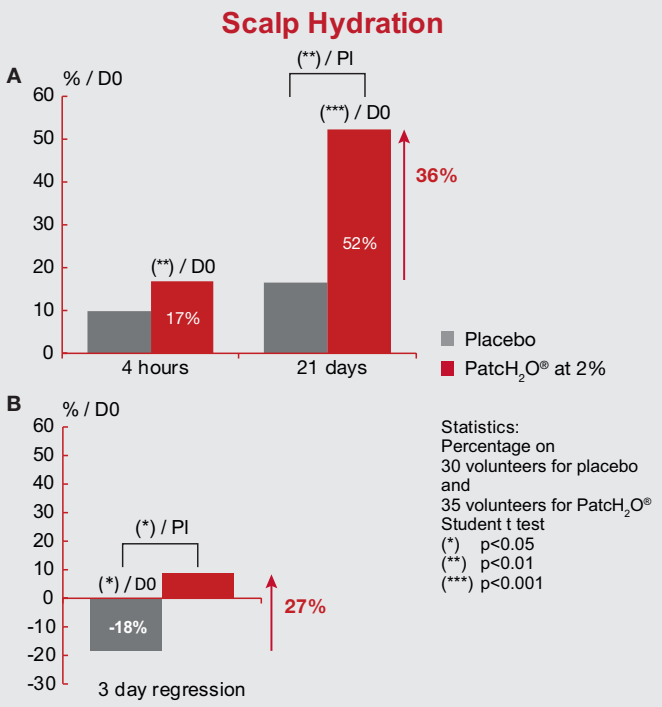


Figure 13 - Hydrating effect on scalp **A** - during treatment and **B**- after stopping the treatment.

CONCLUSION

PatcH$_2$O® treatment is associated with an immediate, cumulative and sustained hydration of dry scalp *in vitro*.

RESULTS & DISCUSSION

Perceptible consumer benefits on scalp comfort and healthy looking hair

After 21 days of product use, PatcH$_2$O® was perceived by a significant majority of the volunteers as reducing itchiness on the scalp (77%) whereas use of placebo formulation did not elicit such response (Figure 14A).

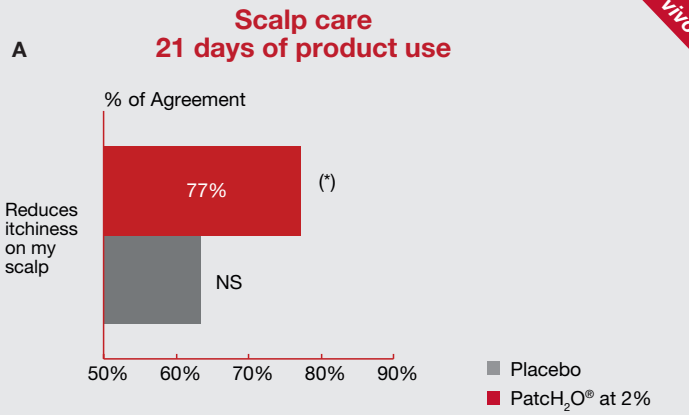
Additional hair benefits were observed by volunteers. After 21 days of product use, PatcH$_2$O® was perceived by a significant majority of the volunteers as making their hair feel stronger (71%), look shiny (80%) and as keeping hair from looking lifeless (71%). These benefits were not observed by volunteers using placebo formulations (Figure 14B).

The effect of PatcH$_2$O® was still perceived three days after the end of treatment. A significant majority of volunteers (69%) felt their scalp more relaxed (Figure 15).

CONCLUSION

Daily treatment with PatcH$_2$O® is associated with the perception of itchness reduction and healthy-looking hair.

in vivo



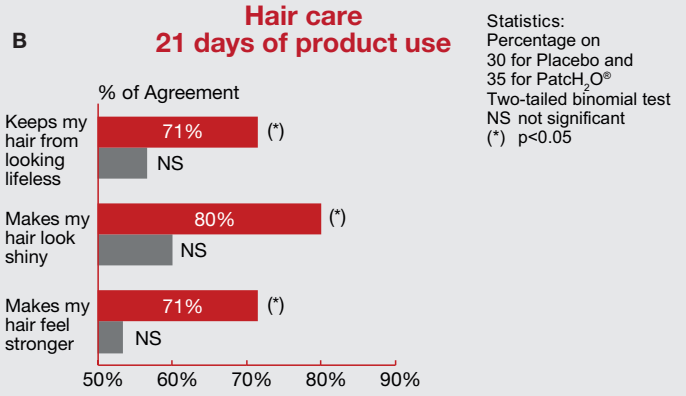


Figure 14 - Self-perception by consumers on **A** scalp comfort and **B** hair quality benefits, after 21 days of daily product application.

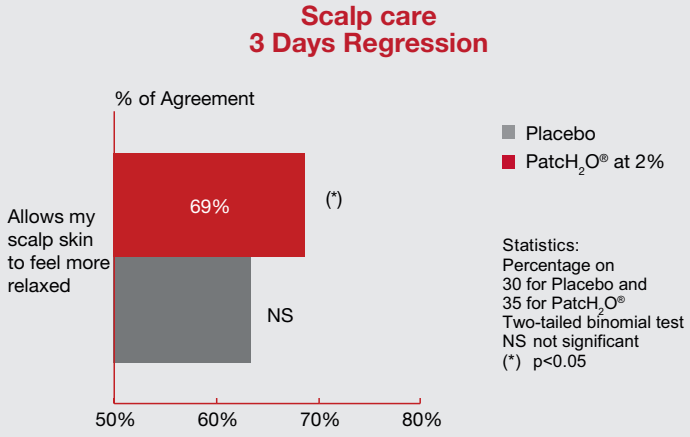


Figure 15 - Self-perception by consumers of retained effect on scalp comfort 3 days after stopping the product application.

MATERIALS&METHODS

Study design

The clinical trial was conducted in a randomized and double-blinded manner to evaluate the effect of PatcH$_2$O® against a placebo. One group of volunteers was treated with a leave-on hair serum formulation containing PatcH$_2$O® at 2% and another group was treated with the same formulation without the active (placebo). The formulations are detailed in Annex 2.

Volunteers were allowed to use their pre-approved shampoo and rinse-off conditioner without any actives during the washout phase and for the duration of the study. No dying, chemical or another scalp treatment were allowed during the study. Volunteers were asked to wash their hair at least 2 hours before the visit to the testing center.

Inclusion criteria

The study was done on 65 healthy female volunteers aged 18-65 years, from different ethnicities (Caucasian, Afro-American, Hispanic and Asian) having dry hair and considering themselves to have an itching/uncomfortable and dry scalp and having a Skicon measurement < 50 µS on the scalp.

Application modality

30 volunteers have applied the placebo hair serum and 35 volunteers have applied the hair serum containing PatcH$_2$O® at 2%. Volunteers applied the product once a day on the full scalp. Each application consisted of 10 full droppers. 1 dropper in the line down the center of the scalp, 3 droppers on the left side of the scalp, 3 droppers on the right side of the scalp and 3 droppers on the back of the scalp. After application, volunteers massaged the product into the scalp thoroughly. Volunteers applied products after baseline measurements till 21 days. Volunteers stopped applying products from the 22nd day still the 24th day (3-day regression).

Evaluation methods

Skicon measurements

The principle of the method is based to the relationship between dielectrically conductance and water content of the skin. Measurements were performed by Skicon device (Skicon 200 EX, I.B.S. Co. Ltd, Japan). Data are reported in micro-Siemens (µS). Skicon measurements have been reported to correlate well to dry skin clinical grades[31] and to *in vivo* measurements of water content by infrared spectroscopy[32]. Before the measurements volunteers were acclimated to controlled environmental conditions for at least 30 minutes. Study rooms will be maintained at 20-30ºC and 45-65% RH with no condensation. Skicon measurements were taken at 9 different locations on the scalp using the 8-pin probe.

Consumer perception

The effectiveness of the product on scalp and hair and overall opinion of experience of consumers using the tested product were evaluated by a self-perception questionnaire. Questionnaires were completed at the test center before any measurement. Volunteers individually complete the questionnaires in front of a mirror with a standardized light and without any extrinsic influences (other volunteers, results of technical measurements). The raw data were collected and subjected to statistical analysis.

Results and statistics

Results from the Skicon are expressed as mean percentage of evolution versus baseline (%/D0). The statistical analysis of the evolution of the Skicon measurements in function of time were done after the verification of the normality of distribution using Shapiro-Wilk test. Afterwards, the statistical analysis of the evolution of the studied parameters for each group was performed with the Student t test in case the normality of distribution was confirmed. In the case of the distribution had not followed the normal law, a non-parametric test (Mann-Whitney test) was used.

Number of positive or negative grouped responses from the self-perception questionnaire were analyzed by two-tailed binomial test based on the frequency of responses[33].

The threshold of significance was set to 5% ($p<0.05$).

in vivo

Conclusion

PatcH$_2$O® on dry scalp and hair - human volunteers

Hydra-comfort scalp: PatcH$_2$O® brings immediate, cumulative and sustained moisture to the dry and itchy scalp to recover scalp comfort and well-being.

- Immediate scalp hydration (+17% over baseline).
- Cumulative scalp hydration after 21 days of treatment (+36% over placebo).
- Sustained hydrating effect on scalp 3 days after stopping the treatment (+27% over placebo).
- 77% of consumers perceived a reduction of scalp itchiness after 21 days of treatment.
- 69% of consumers perceived their scalp more relaxed 3 days after stopping the product application

Healthy-looking hair: PatcH$_2$O® rebalance hair moisture of dry hair to recover shine, strength and vitality. After 21 days of daily application:

- 80% of consumers perceived their hair look shiny.
- 71% of consumers perceived their hair stronger.
- 71% of consumers perceived their hair less lifeless.



GENERAL CONCLUSION

Consumers need a solution to rehydrate and protect scalp and hair, that has been dried out by external factors and hair grooming habits.

Dry scalp is often uncomfortable and itchy. The main reasons for dry scalp are underlying health conditions and frequent and harsh treatments linked to hair grooming habits. These concerns worsen during winter due to cold weather and wind, or during summer due to hot and dry air conditions. Dry scalp can cause frizzy and dry hair. The right way to take care of dry scalp is to retain the moisture in the scalp.

When hair does not retain enough moisture, it turns dry. Dry hair is porous and exhibits signs of damage. Damaged hair is less hydrophobic than virgin hair and absorbed more water during aqueous treatment. The porous structure of damaged hair leads to higher loss of water by evaporation which is exacerbated in the drying period in comparison to healthy hair. Lack of water reduces shine making hair appears frizzy, dull and lifeless.

PatchH_2O®, a 3D molecular network of biopolymers, performs the controlled release of a moisturizing complex on both scalp and hair.

PatchH_2O® is a water soluble product that forms a protective film at the surface of scalp and hair fibers thanks to its 3D network of bio-polymers (hyaluronic acid, alginate and pullulan). This film-forming structure acts as a barrier to fight against water loss by evaporation. In addition, PatchH_2O® contains highly concentrated hydrating substances (trehalose, glycerin, urea and L-serine) which allow rehydration of scalp and hair. The combination of the micro-patch of biopolymers and the moisturizing complex provides a sustained and effective hydrating effect due to a controlled release mechanism.

By using PatchH_2O®, dry and thirsty scalp is effectively hydrated and protected against the excessive loss of water. Scalp feels more comfortable and relaxed.

By using PatchH_2O®, dry and dull hair turn into healthy looking hair. Hair is perceived shinier, stronger and is back to life.

ANNEX 1

Technical data - Available upon request

- Quality and regulatory product information
- Composition sheet
- Specifications
- Formulation data sheet
- Marketing and scientific brochures for skin care

ANNEX 2

Clinical test formula

Trade name	INCI name	Placebo formulation %	PatcH$_2$O® formulation %
Water	Water	96.05	94.05
Elestab™ 388	Propylene Glycol, Phenoxyethanol, Chlorphenesin, Methylparaben	2.50	2.50
Glycerin	Glycerin	1.00	1.00
Rheocare® XG	Xanthan Gum	0.20	0.20
Cosmedia® SP	Sodium Polyacrylate	0.25	0.25
PatcH$_2$O® A00297	Water, Glycerin, Glyceryl Polyacrylate, Trehalose Urea, L-serine, Pentylene Glycol, Algin, Caprylyl Glycol, Sodium Hyaluronate, Pullulan, Disodium Phosphate, Potassium Phosphate	-	2.00

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ANNEX 3

Formulation examples

Hydra-comfort scalp - Sulfate free shampoo (rinse-off)

Gentle shampoo to remove dirt, pollution and product build up. Leaves on scalp and hair feeling of cleanliness but not stripping. Silicones, sulfates and cationic products could increase the dryness of scalp and could build-up effect on hair. Gentle and hydrating cleansers soothe dry and uncomfortable scalp and prevent split ends and other types of hair damage. Bring a touch of wellbeing for a cocoon therapy - HB-FR-16-BC-50777/02.

Phase	Ingredients	INCI	% by weight	Function
A	Water, demin.	Aqua	60.30	
	Rheocare® XG (Europe only)	Xanthan Gum	1.20	Rheology modifier
B	Plantacare® 2000 UP	Decyl Glucoside	14.00	Surfactant
	Plantasil® Micro	Dicaprylyl Ether, Decyl Glucoside, Glyceryl Oleate	5.00	Conditioning agent
	Plantapon® ACG HC	Sodium Cocoyl Glutamate	12.00	Surfactant
	Lamesoft® PO 65	Coco-Glucoside, Glyceryl Oleate	2.00	Conditioning agent
	Glycerin	Glycerin	3.00	Stabilizer
	Preservative		q.s.	Preservative
C	Citric Acid (50% solution)	Citric Acid	q.s.	pH Adjustment
D	Perfume	Parfum	0.50	Fragrance
	PatcH2O™ A00297	Aqua, Glycerin, Glyceryl Polyacrylate, Trehalose, Urea, Serine, Pentylene Glycol, Algin, Caprylyl Glycol, Sodium Hyaluronate, Pullulan, Disodium Phosphate, Potassium Phosphate	2.00	Active ingredient

Specifications	
pH value (23°C)	5.2
Viscosity (Brookfield; RVT; spindle 5; 50 rpm; 23°C)	2 200 mPas

Performance

Additional performance has not been evaluated

Manufacturing Process

Phase A : At room temperature, disperse Rheocare® XG in water until the gel is homogeneous.
Add one by one the components of Phase B.
Adjust pH to 4.8 - 5.3 with Citric Acid 50 % (Phase C).
Finally add components of Phase D.

Formulation examples

Post-hair dye/straightening - Soothing serum (leave-on)

Hydrates and soothes scalp immediately and durably with a light and well-accepted formula. Re-balance water content on dry and itchy scalp. Also gives to hair the power to trap and retain moisture. It is a new ritual to take daily care of scalp and hair, especially after dying or straightening procedures - SC-FR-17-BC-50751/02.

Phase	Ingredients	INCI	% by weight	Function
A	Water, demin.	Aqua	96.55	
	Preservative*		q.s.	Preservative
	Glycerin	Glycerin	1.00	Humectant
B	Rheocare® XG (Europe only)	Xanthan Gum	0.20	Stabilizer
	Cosmedia® SP	Sodium Polyacrylate	0.25	Rheology modifier
C	PatcH2O™ A00297	Aqua, Glycerin, Glyceryl Polyacrylate, Trehalose, Urea, Serine, Pentylene Glycol, Algin, Caprylyl Glycol, Sodium Hyaluronate, Pullulan, Disodium Phosphate, Potassium Phosphate	2.00	Active ingredient

Specifications	
pH value (23°C)	6.2
Viscosity (Brookfield; RVT; spindle 5; 50 rpm; 23°C)	1200 mPas

Performance

Additional performance has not been evaluated

Manufacturing Process

1- Mix ingredients of A homogeneously at 40°C. Cool at room temperature.
2- Add Phase B into A while stirring.
3- When phase A+B is homogenesous, add phase C under stirring.

Stability Tests

Stable 3 months at 4°C, RT, 40°C, 45°C

Formulation examples

Healthy look - Conditioner for scalp & hair (rinse-off)

Healthy scalp for healthy looking hair. Saves the scalp from dryness and keeps the healthy look of the hair. Environment-friendly by using biodegradable and solely based on renewable raw materials. Produces a feeling of security and comfort. Get back your shiny and beauty hair thanks to a perfect hair'dration - HB-FR-16-BC-50772-03.

Phase	Ingredients	INCI	% by weight	Function
A	Water, demin.	Aqua	88.17	
	Plantaquat® NC	Cetearyl Alcohol, Lecithin, Sodium Cetearyl Sulfate, Olus Oil [EU], Cetearyl Alcohol, Lecithin, Sodium Cetearyl Sulfate, Vegetable Oil [CTFA]	7.00	Conditioning agent
B	Myritol® 312	Caprylic/Capric Triglyceride	2.00	Emollient
C	Preservative		q.s.	Preservative
D	Citric Acid	Citric Acid	0.33	pH Adjustment
E	PatcH2O™ A00297	Aqua, Glycerin, Glyceryl Polyacrylate, Trehalose, Urea, Serine, Pentylene Glycol, Algin, Caprylyl Glycol, Sodium Hyaluronate, Pullulan, Disodium Phosphate, Potassium Phosphate	2.00	Active ingredient
	Perfume	Parfum	0.50	Fragrance

Specifications

pH value (23°C)	4.35
Viscosity (Brookfield; RVT; spindle TD, helipath; 5 rpm; 23°C)	14 000 mPas

Performance

Additional performance has not been evaluated

Manufacturing Process

Heat water to 70-75 °C. Add Plantaquat® NC while stirring.
Add phase B at 50-55 °C and homogenise.
Add phase C at 30 °C and adjust pH using phase D.
Add phase E while stirring.

Additional Information

Perfume Technicoflor (Cocoa Butter RS61518)

Stability Tests

Conform 3 months at 4°C, RT, 45°C
No microbiological testing carried out.

Formulation examples

Butter-to-oil for ethnic care - Ultra-moisture conditioner (leave-on)

Experience the nice changing texture of the butter-to-oil for easy and pleasant application. Gives ultra-moisture for the driest hair - HB-FR-16-BC-50776-02.

Phase	Ingredients	INCI	% by weight	Function
A	Water, demin.	Aqua	47.00	
	Glycerin	Glycerin	5.00	Humectant
	1,2 Butanediol *(Interchimie)*	Butylene Glycol	3.00	Stabilizer
	Preservative		q.s.	Preservative
B	Eumulgin® BA 25	Beheneth-25	3.00	Emulsifier (O/W)
	Lanette® 22	Behenyl Alcohol	2.00	Consistency agent
	Cetiol® SB 45	Butyrospermum Parkii Butter	12.00	Emollient
	Cetiol® RLF	Caprylyl Caprylate/ Caprate	7.00	Emollient
	Cetiol® Sensoft	Propylheptyl Caprylate	10.00	Emollient
	Cosmedia® SP	Sodium Polyacrylate	1.50	Rheology modifier
C	Cetiol® Ultimate	Undecane, Tridecane	7.00	Emollient
D	PatcH2O™ A00297	Aqua, Glycerin, Glyceryl Polyacrylate, Trehalose, Urea, Serine, Pentylene Glycol, Algin, Caprylyl Glycol, Sodium Hyaluronate, Pullulan, Disodium Phosphate, Potassium Phosphate	2.00	Active ingredient
	Perfume	Parfum	0.50	Fragrance

Specifications	
pH value (23°C)	5.85
Viscosity (Brookfield; RVT; spindle TD, helipath; 5 rpm; 20°C)	152 000 mPas

Performance

Additional performance has not been evaluated

Manufacturing Process

Heat phases A and B at 80°C.
Introduce phase B in phase A and have high stir during 10 minutes.
Cool down at 60°C, then introduce phase C and have a high stir during 5 minutes.
Cool down at room temperature and finally add phase D.



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